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ABOUT ME

Results-driven **Machine Learning and Researcher** with 3+ years of experience in **computer vision, deep learning, and applied AI** across academic research and industrial production. Expertise in **OCT, image and signal processing, PyTorch**, and end-to-end AI development. Author of **5 peer-reviewed journal papers**, including **4 as co-first author** and **4 in Q1 journals**. Experienced in **rigorous evaluation, reinforcement-learning environments**, and deploying reliable computer-vision systems in **real-world production**.

SKILLS

Programming

C++ / Python

Machine Learning Frameworks

PyTorch / OpenCV / NumPy / Scikit-Learn / TensorFlow / Hugging Face / Pandas

Technical

Image & Signal Processing / Machine Learning / Deep Learning / Generative AI / Transformers / Computer Vision / Reinforcement Learning / Optical Coherence Tomography (OCT)

WORK EXPERIENCE

EVOQ Systems – Colombo, Sri Lanka

Machine Learning Engineer

[01/12/2024 – Current]

- Developed and deployed an end-to-end anomaly detection system for inspecting **4,500+ heat-sealed garment logos/day across 3 production units** at Sri Lanka's largest apparel manufacturer, achieving **>95% accuracy** through OpenCV-based boundary analysis and deep feature reconstruction using **PyTorch**.
- Designed and deployed a real-time seam-defect detection system for a leading apparel manufacturer, detecting micro-snags with **>95% accuracy** using lightweight classical computer vision and machine-learning techniques for **CPU-only inference**.
- Engineered a novel **few-shot defect-localization** method for detecting micro-defects in rubber, dipping, and electrical gloves, achieving reliable localization from only **20-25 annotated samples** through feature-matching techniques implemented in PyTorch, overcoming severe data scarcity.
- Developed a **dual-camera computer-vision system** for automated fabric-roll width measurement, achieving **>98% accuracy** using OpenCV-based detection and homography-based image-to-real-world coordinate mapping.
- Contributed to the development of a **machine-learning model for ATM cash-replenishment forecasting**, achieving **>80% forecasting accuracy** and supporting the reduction of idle-cash and cash-in-transit (CIT) costs for a Sri Lankan banking deployment.

- Leading applied research on **Vision-Language Models (VLMs)** for real-time CCTV surveillance, including local deployment and optimization of **Qwen-VL**, while developing a generative-AI pipeline for rendering kitchen designs from natural-language customer descriptions.
- Developing and evaluating **LiDAR point-cloud processing pipelines** involving filtering, denoising, downsampling, alignment, reconstruction, and quantitative 3D analysis.

 **Thullex** – Berlin, Germany

Computer Vision Engineer (Remote Consultant)

[01/05/2024 – Current]

- Led the modernization of the company's core computer-vision measurement technology, replacing legacy image-processing pipelines based on contour detection and color segmentation with **deep-learning approaches including RF-DETR, U-Net, keypoint detection, and object detection**, now powering production measurement systems.
- Architected and deployed an end-to-end deep-learning garment-measurement system combining **RF-DETR-based detection/segmentation** with a novel homography-based image-to-real-world coordinate transformation, achieving **>98% accuracy within ± 3 mm tolerance**. Deployed **4 production units** at one of Thailand's largest apparel manufacturers, each processing **~200 garments/day**, with **<2 s CPU inference** enabled by INT8 post-training quantization using OpenVINO.
- Designed and deployed the detection architecture for an automated baby-bodysuit measurement system, achieving **± 3 mm measurement accuracy** and enabling production deployment at a leading Sri Lankan apparel manufacturer.
- Extended the deep-learning measurement platform to **undergarment measurement**, adapting the existing detection architecture to support an additional production line and garment category.

 **MAS Linea Aqua** – Colombo, Sri Lanka

Undergraduate Researcher / R&D Intern

[01/08/2023 – 01/01/2024]

- Architected and developed the complete computer-vision and software stack for an **Intelligent Ultrasonic Tacking robotic system**, enabling **real-time sub-pixel localization** on 2D textile surfaces; now deployed in production and forming the basis of a **Q1 journal publication** in *Results in Engineering* (Elsevier).
- Developed the **MUFF-TM (Multichannel Feature-Fusion Template Matching) method**, enabling robust and precise real-time part localization for automated robotic tacking under challenging textile conditions.
- Integrated the vision system with **industrial automation infrastructure**, connecting AI-based localization with **PLCs, robotic systems, and hardware interfaces** to deliver a complete production-ready automation solution.

PUBLICATIONS

[2026]
[Vision-incorporated Multichannel Feature Fusion Template Matching \(MUFF-TM\) and real-time sub-pixel coordinate localization for 2D textile surfaces in ultrasonic tacking systems](#) Q1 journal (IF ≈ 9.4), Open Access. Co-first author. Computer-vision method with real-time sub-pixel localization; basis of an industrial ultrasonic-tacking system deployed in production.

Journal Name: Results in Engineering | **Volume, Issue and Pages:** Vol. 31, Article 111491 | **Publisher:** Elsevier
 Kahandawala B.S.†, Nalmi R.†, Sodige B.A.K., Subasinghage K., Silva B.N., Wijesinghe R.E., Woo S.T. (2026). Vision-incorporated Multichannel Feature Fusion Template Matching (MUFF-TM)... Results in Engineering, Vol. 31, 111491.

[2026]
[OcupHI: knowledge-driven colorimetric interpretation framework for high-precision real-time ocular pH diagnostics](#) Q2 journal. Co-first author. Real-time colorimetric framework for biomedical (ocular pH) diagnostics.

Journal Name: Emergent Materials | **Volume, Issue and Pages:** Vol. 9, Article 204 | **Publisher:** Springer Nature

Kahandawala B.S.†, Sandaruwan H.H.P.B., Liyanage P., Dassanayake R.S., Costha N.P., Liyanage R.N., Wijenayake U., Wijesinghe R.E., Silva B.N., Manatunga D.C. (2026). OcupHI... Emergent Materials, Vol. 9, Article 204.

[2026]

Brewing plastics: OCT reveals microplastic release from nylon tea bags in simulated brewed tea infusions Q1 journal. Co-first author. OCT-based detection of microplastics; featured in national press.

Journal Name: Environmental Science: Processes & Impacts | **Volume, Issue and Pages:** Vol. 28, Issue 2, pp. 392–404 | **Publisher:** Royal Society of Chemistry (RSC)

Jayasekara P.M.†, Abhishek P.†, Kahandawala B.S.†, et al. (2026). Brewing plastics: OCT reveals microplastic release from nylon tea bags. Environmental Science: Processes & Impacts, 28(2), 392–404.

[2026]

High-resolution optical imaging for sustainable fish freshness and safety assessment

Q1 journal. Co-first author. High-resolution OCT imaging for non-destructive food-quality assessment.

Journal Name: Optik | **Volume, Issue and Pages:** 2026 (Article 172733) | **Publisher:** Elsevier

Madhubhashini M.N.†, Kahandawala B.S.†, Sandaruwan H.H.P.B., Silva B.N., Wijenayake U., Wijesinghe R.E. (2026). High-resolution optical imaging for sustainable fish freshness and safety assessment. Optik, 172733.

[2025]

Environmental forensics of the X-Press Pearl disaster: uncovering the internal micro-structural transformations in marine microplastics Q1 journal (IF ≈ 13.6). Co-author. OCT-based environmental forensics of marine microplastics.

Journal Name: Journal of Hazardous Materials | **Volume, Issue and Pages:** Vol. 496, Article 139231 | **Publisher:** Elsevier

Jayasekara P.M., Abhishek P., et al. (2025). Environmental forensics of the X-Press Pearl disaster: micro-structural transformations in marine microplastics. J. Hazardous Materials, 496, 139231.

EDUCATION AND TRAINING

Master of Science (MSc) in Artificial Intelligence

Sri Lanka Institute of Information Technology (SLIIT) [01/02/2024 – Current]

City: Colombo | **Country:** Sri Lanka | **Level in EQF:** EQF level 7

Currently pursuing an MSc in Artificial Intelligence (by Research), focusing on computer vision, deep learning, and applied AI. Active researcher at the Faculty of Computing (CIET), SLIIT. Expected completion: 2028.

Bachelor of Engineering Technology Honours (Hons) in Mechatronics Engineering

University of Sri Jayewardenepura [01/04/2020 – 28/04/2024]

City: Colombo | **Country:** Sri Lanka | **Level in EQF:** EQF level 6

Graduated with First Class Honours (GPA 3.74 / 4.0)

LANGUAGE SKILLS

Other language(s):

English

LISTENING C1 **READING** C1 **WRITING** C1

SPOKEN PRODUCTION C1 **SPOKEN INTERACTION** B2

Levels: A1 and A2: Basic user; B1 and B2: Independent user; C1 and C2: Proficient user